Technical Debt in Latin American Academic Software: A Matched-Sample Comparison with Open-Source Controls

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Abstract

Software produced in university courses and capstone projects is widely assumed to carry more technical debt than comparable open-source work, but this has rarely been tested for Latin America. We compare 274 academic GitHub repositories (identified by thesis and course keywords, with contributors located in Latin America) against 275 non-academic repositories matched by language and size on nine static metrics, using Mann-Whitney U tests with Holm correction and Cliff's delta. Contrary to our hypothesis, academic code shows *significantly lower* cyclomatic complexity (median 2.16 vs. 3.12, $\delta = -0.37$, $p < 10^{-11}$). Inheritance depth is higher in academic Java code ($\delta = +0.29$, $p = 0.003$); coupling, cohesion, duplication, and test-file presence do not differ significantly. The most likely explanation is the control sample, whose small, non-labeled repositories resemble learning-stage code more than mature open-source projects. SonarQube on a 415-repository subset corroborates the complexity result ($\delta = -0.458$, $p < 10^{-14}$) and finds a slightly lower technical debt ratio in academic code ($\delta = -0.142$, $p = 0.025$). Self-admitted technical debt markers run in the hypothesized direction: 42.2% of academic repositories contain at least one, against 23.0% of controls ($\delta = +0.190$, $p < 0.001$).

Keywords: Technical debt, software engineering education, empirical software engineering, Latin American software, code complexity, GitHub mining.

Resumen. Se suele suponer que el software producido en cursos universitarios y trabajos de titulación acumula más deuda técnica que proyectos de código abierto comparables, pero esta suposición casi no se ha contrastado para América Latina. Comparamos 274 repositorios académicos de GitHub (identificados por palabras clave de tesis y cursos, con contribuyentes ubicados en América Latina) con 275 repositorios no académicos emparejados por lenguaje y tamaño, en nueve métricas estáticas, mediante pruebas U de Mann-Whitney con corrección de Holm y delta de Cliff. Contra nuestra hipótesis, el código académico presenta una complejidad ciclomática significativamente *menor* (mediana 2,16 frente a 3,12; $\delta = -0{,}37$; $p < 10^{-11}$). La profundidad de herencia es mayor en el código académico en Java ($\delta = +0{,}29$; $p = 0{,}003$); acoplamiento, cohesión, duplicación y presencia de archivos de prueba no difieren significativamente. La explicación más probable está en la muestra de control, cuyos repositorios pequeños y sin etiquetar se parecen más a código de aprendizaje que a proyectos maduros. SonarQube, aplicado a 415 repositorios, corrobora el resultado de complejidad ($\delta = -0{,}458$; $p < 10^{-14}$) y encuentra un ratio de deuda técnica ligeramente menor en el código académico ($\delta = -0{,}142$; $p = 0{,}025$). Los marcadores de deuda técnica autoadmitida siguen la dirección de la hipótesis: el 42,2 % de los repositorios académicos contiene al menos uno, frente al 23,0 % del grupo de control ($\delta = +0{,}190$; $p < 0{,}001$).

Palabras clave: Deuda técnica, educación en ingeniería de software, ingeniería de software empírica, software latinoamericano, complejidad de código, minería de GitHub.

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1 Introduction

It is commonly assumed, among educators and practitioners alike, that code written for a course assignment or a thesis project carries more technical debt, in the sense of the metaphor introduced by Cunningham [1] (higher complexity, near-zero test coverage, more duplicated code), than comparable work produced outside an academic setting. This assumption motivates a large share of software-engineering curriculum design, yet it is rarely tested with a matched, quantitative comparison, and to our knowledge no such comparison has been run specifically for Latin American academic software, where institutional resourcing, course structure, and open-source contribution culture may all differ from the North American and European settings most existing studies draw on.

This paper runs that comparison directly. We ask whether repositories produced in Latin American university courses and capstone/thesis projects differ measurably, on a set of standard static metrics, from a size- and language-matched sample of non-academic repositories, and if so, in which direction and by how much. We report the result even where it contradicts our starting hypothesis, because a null or reversed finding here is itself informative about what "academic code" means empirically in this region, as opposed to what it is assumed to mean.

The contributions of this work are: (1) a size- and language-matched quantitative comparison of Latin American academic software against non-academic controls, at a scale (549 repositories) sufficient for non-parametric hypothesis testing; (2) an honest report of an inverted finding, lower rather than higher complexity in academic code, together with a specific, testable account of why (control-group composition) rather than a reframing of the study to fit the original hypothesis; (3) a cross-tool robustness check of the central finding using SonarQube Community itself, run without Docker from its standalone distribution, which corroborates the complexity result across distinct parser implementations and surfaces a genuine tool-dependent discrepancy in duplication rate that we report rather than resolve.

2 Related Work

An early in-depth exploration of technical debt by Tom, Aurum, and Vidgen [2] combined a review of the literature with practitioner interviews to describe the dimensions of technical debt (among them code, design and architecture, testing, and documentation debt), why it arises, and its consequences; this vocabulary frames the kinds of debt our metrics and SATD markers can and cannot capture. Li, Avgeriou, and Liang [3] subsequently mapped 94 studies on technical debt and its management, classified TD into ten types, and found that code-related debt has received the most attention, while calling for more empirical studies with high-quality evidence. We are not aware of a prior study that compares academic and non-academic code on these metrics, controlled by language and size, at the scale reported here.

Santos et al. [4] conduct a systematic literature review of external quality metrics for object-oriented software; we apply metrics of the coupling, cohesion, and complexity families they review to a labelled academic/non-academic comparison. More directly relevant, Hamer et al. [5] use Git-derived metrics to measure students' and teams' code-contribution patterns within university software-development courses, across eight projects with 150 students, showing that repository-level metrics can characterize academic software engineering practice; our study extends this direction from process metrics (who contributed what) to product metrics (what the resulting code looks like structurally), and from within-course comparisons to a labeled comparison against non-academic repositories. Mateos et al. [6] apply the same Chidamber and Kemerer [7] object-oriented metric suite this paper uses, in their case to relate service-interface complexity to the underlying object-oriented implementation; their use of CK metrics as an early, development-time complexity signal is consistent with how this paper treats complexity metrics generally, as a proxy available before a system reaches production rather than as an after-the-fact quality judgment. The repository-mining methodology follows the practices established by Kalliamvakou et al. [8] and refined by Munaiah et al. [9], both of which show that many GitHub repositories are personal, inactive, or not engineered software, so samples need filtering before structural metrics can be read as representative of software engineering practice; our fork exclusion and 10-commit minimum follow that advice in a simplified form. Munaiah et al. explicitly list homework assignments as an example of the non-engineered repositories that such filtering removes, which is a reminder that our academic sample is, by design, drawn from the kind of repository that engineered-project studies exclude.

The self-admitted technical debt analysis in Section 3.4 is grounded in a body of work that began with Potdar and Shihab [10], who identified SATD in the source-code comments of four large open-source projects (Eclipse, Chromium OS, Apache HTTP Server, and ArgoUML), found it in 2.4% to 31% of files, and reported that more experienced developers introduce most of it, while time pressure and code complexity do not correlate with how much SATD is present; their figures are per file, so they are not directly comparable

with our repository-level rates (42.2% academic vs. 23.0% control). Bavota and Russo [11] extend this line of work to a large-scale study of how SATD spreads and evolves across many open-source projects. Maldonado and Shihab [12] read more than 33,000 comments and classify SATD into five types (design, defect, documentation, requirement, and test debt), with design debt the most common; our marker list (`TODO`, `FIXME`, `HACK`, `XXX`, `BUG`) is a simpler keyword approximation of the comments they analyse manually. Huang et al. [13] subsequently detect SATD in source comments automatically with a text-mining approach that combines feature selection with a composite classifier trained on several projects; we use the simpler keyword approach because precision is less critical for our research question (is the *prevalence* of any SATD marker different between academic and control repositories?) than for studies aimed at classifying SATD type. Alfayez et al. [14] analyse 128,897 technical-debt-related posts on Twitter and find growing interest in the topic, grouped into 11 categories; we cite it as evidence that technical debt is widely discussed in practice, beyond the research literature.

For the SonarQube corroboration subsection, Amanatidis et al. [15] build an empirical benchmark to evaluate how far different technical-debt measurement tools (including SonarQube) agree with each other, showing that the estimate obtained depends on the tool; this contextualises the discrepancy we observe between `jscpd` and SonarQube's duplication detectors as an instance of a documented pattern rather than an anomaly specific to our sample or setup. Lenarduzzi et al. [16] use data-driven regression models to estimate the additional project cost of sub-optimal code conditions and report that SonarQube's debt estimates can be made more precise; their work suggests both that automated TD estimation is an active research problem and that the tool's absolute figures are a starting point rather than a ground truth, two assumptions that underpin our decision to use SonarQube as a corroboration instrument rather than as the primary measurement.

3 Methodology

3.1 Sample construction

Candidate academic repositories were identified via the GitHub search API using Spanish/Portuguese academic keywords (*tesis*, *trabajo de titulación*, *proyecto integrador*, *trabalho de conclusão*, *tcc*, and related terms) combined with a language filter (Java, Python, JavaScript) and a minimum size threshold, then verified by checking whether at least one of the repository's top contributors declares a Latin American city or country in their public GitHub profile location field. This combined keyword-plus-location approach is a heuristic, not a ground-truth label, and is discussed as a limitation below. Repositories with fewer than 10 commits were excluded. The GitHub search API excludes forked repositories by default unless a `fork:` qualifier is explicitly added, and none of our queries added one; we verified this directly by re-issuing representative academic- and control-sample queries and confirming `fork=false` for every returned repository, so the sample does not require a separate fork-filtering step. From 5,418 unique keyword-matched candidates, 300 were accepted as academic (274 successfully cloned and analyzed; the remainder had since been deleted, renamed, or were otherwise unreachable). A control sample of 300 non-academic repositories was drawn by querying GitHub for repositories in the same three languages within a size band matched to the academic sample (275 successfully analyzed), explicitly *not* sorted by popularity, since an earlier version of this pipeline that sorted by star count pulled in large, well-known projects that violate the intended size-matched pairing. We refer to this design as a matched-sample comparison because matching was performed at the group level (by language and size band) rather than pairing repositories one-to-one, and independent-samples statistical tests are used accordingly.

Table 1 reports the country and language composition of the 300 academic repositories, resolved from the GitHub profile location of each repository's top contributors (the same field used for the LatAm-verification heuristic above). Brazil accounts for nearly half the sample (47.7%), consistent with Brazil's outsized share of Latin American GitHub activity; the remaining sample spans 15 further countries (16 in total).

3.2 Metrics

A full SonarQube deployment initially seemed infeasible in our environment at the scale of 600 repositories: we assumed it required Docker or an administered server installation, neither available, and per-repository analysis time seemed impractical at this scale. We substituted a combination of lightweight, license-compatible tools that jointly cover the target empirical metric suite: `lizard` for cross-language cyclomatic complexity [17] and lines of code, `jscpd` for cross-language code duplication, and `ck` (a standard standalone reference tool that runs from a single `.jar` without requiring a SonarQube server) for the Chidamber and Kemerer [7] object-oriented metrics suite on the Java subset (coupling between objects, CBO; lack of cohesion, LCOM; depth of inheritance tree, DIT; weighted methods per class, WMC). Test coverage

Table 1: Composition of the initially accepted academic sample ($n = 300$) by country and language

Country	n	%	Java	JavaScript	Python	Analyzed*
Brazil	143	47.7	57	42	44	130
Argentina	50	16.7	15	25	10	46
Colombia	33	11.0	14	9	10	30
Peru	17	5.7	7	8	2	15
Ecuador	13	4.3	8	2	3	12
Mexico	12	4.0	1	8	3	11
Chile	8	2.7	3	1	4	7
Venezuela	5	1.7	0	4	1	5
El Salvador	4	1.3	1	2	1	4
Uruguay	4	1.3	2	2	0	4
Costa Rica	3	1.0	1	1	1	3
Other	7	2.3	1	4	2	6
Unresolved	1	0.3	1	0	0	1

Other: Bolivia, Cuba, Guatemala, Honduras, Panama; each with fewer than 3 repositories. *26 repositories (8.7%) were unavailable at clone time; the analyzed subset ($n = 274$) retains approximately this proportional distribution.

was not measured by executing test suites, which is infeasible across 600 arbitrary repositories without individualized build configuration; instead we report the proportion of source files that are test files by naming convention, an explicit proxy rather than measured coverage.

3.3 SonarQube corroboration

During methodological refinement and extended validation, we revisited the assumption above: SonarQube Community can in fact run without Docker, from a standalone ZIP distribution, as a plain Java application (Java 21, provisioned locally, no administrator privileges required). We installed it this way and re-cloned the full sample from GitHub. The original clones had been deleted immediately after metric extraction to save disk space, so no source code remained on disk to point SonarQube at; we succeeded in re-cloning 546 of the original 600 repositories (274 academic, 272 control; a handful had since been deleted, renamed, or made private). SonarQube's Java analyzer refuses to run its full rule set on source files without matching compiled bytecode (`sonar.java.binaries`); since compiling 300-plus arbitrary student and hobby projects is infeasible without per-project build configuration, we pointed this property at an empty directory, which lets analysis proceed with Java-bytecode-dependent rules disabled while complexity, duplication, and code-smell detection remain intact. Under this configuration, 415 of the 546 re-cloned repositories (223 academic, 192 control) completed analysis successfully within a 150-second per-repository budget; the rest failed on genuinely large repositories, unsupported content, or transient errors.

We treat this SonarQube-derived sample as a cross-tool consistency check on the `lizard`/`ck`-based findings above, not a replacement for them (it is smaller, and not repository-for-repository identical to the primary sample), comparing groups with the same Mann-Whitney/Holm/Cliff's-delta procedure on SonarQube-native metrics: cyclomatic and cognitive complexity per function, `duplicated_lines_density`, `sqale_debt_ratio` (SonarQube's own technical-debt-ratio metric, the specific metric our original substitution could not reproduce), and code smells per 1,000 lines.

3.4 Statistical analysis

Because software metrics are typically strongly right-skewed, we compare groups with the Mann-Whitney U test on each of nine metrics, apply the Holm correction for multiple comparisons, and report Cliff's delta [18] as a non-parametric effect size, following standard practice for this kind of comparison in the empirical software engineering literature.

4 Results

To clarify the attrition across analysis stages, Table 2 summarizes the sample size available for each comparison. The primary analysis evaluates the 549 repositories that completed the initial clone and metric-extraction step.

Table 3 summarizes all nine metrics for the primary sample.

Table 2: Sample flow and attrition

Stage	Repositories
Candidates (by keyword)	5,418
Accepted as academic	300
Control sample	300
Successfully cloned and analyzed (Primary)	**549** (274 academic + 275 control)
Re-cloned for SonarQube	546 (274 academic + 272 control)
Completed SonarQube analysis (Sub-sample)	**415** (223 academic + 192 control)

Table 3: Academic vs. control comparison, nine metrics (Holm-corrected). IQR = interquartile range (Q1–Q3)

Metric	Median (acad.)	IQR (acad.)	Median (ctrl.)	IQR (ctrl.)	Cliff's δ	Effect	p_{Holm}
Cyclomatic complexity (mean)	2.16	1.66–3.06	3.12	2.17–4.29	−0.37	medium	< 0.001
Cyclomatic complexity (max)	15.5	9.0–28.0	26.0	15.0–46.0	−0.31	small	< 0.001
DIT (inheritance depth, Java)	1.21	1.06–1.41	1.07	1.00–1.28	+0.29	small	0.003
Lines of code	1,592	592–4,033	2,092	593–6,800	−0.10	negligible	0.322
CBO (coupling, Java)	5.74	3.46–7.12	5.17	1.25–7.56	+0.15	small	0.350
LCOM (cohesion, Java)	8.30	2.61–17.10	7.19	0.46–18.50	+0.12	negligible	0.562
WMC (complexity/class, Java)	5.92	4.34–9.35	6.50	5.04–9.84	−0.08	negligible	0.926
Duplication (%)	5.51	1.96–11.54	5.05	1.26–14.06	+0.03	negligible	1.000
Test-file proportion	0.00	0.00–0.06	0.00	0.00–0.10	−0.01	negligible	1.000

Three of the nine metrics remain significant after Holm correction (Figure 1): the two cyclomatic-complexity measures and inheritance depth. All three concern complexity or design structure, but only inheritance depth moves in the hypothesized direction. Cyclomatic complexity is *significantly lower* in academic repositories than in the control group, both on average (Cliff's $\delta = -0.37$, a medium effect) and at the maximum observed per repository ($\delta = -0.31$, small), directly contradicting our initial hypothesis that academic code would be more complex. Inheritance depth (DIT, Java subset only) is significantly higher in academic code ($\delta = +0.29$, small), consistent with the hypothesized pattern of less experienced object-oriented design. Coupling (CBO) trends in the hypothesized direction but does not survive correction ($p = 0.350$). Duplication and test-file proportion show essentially no difference between groups ($\delta \approx 0.03$ and -0.01 respectively), and neither duplication nor size-driven variation in it separates the two groups (Figure 3); notably, the median test-file proportion is *zero in both groups* (Figure 2), meaning near-absent testing is not a distinguishing feature of academic code in this sample; it characterizes both groups about equally.

Because the control group is matched to the academic sample by language as well as by size, and cyclomatic complexity itself varies systematically by programming language, we checked directly whether the lower-complexity finding is an artifact of language composition rather than a genuine academic-versus-control difference. Table 4 repeats the cyclomatic-complexity comparison separately within each of the three languages. The effect holds, and remains significant after a separate Holm correction across the three within-language tests, in all three: Java ($\delta = -0.41$), JavaScript ($\delta = -0.33$), and Python ($\delta = -0.49$), each with medium-to-large effect size. Language composition does not explain the aggregate result.

Table 4: Cyclomatic complexity (mean), academic vs. control, stratified by language (Holm-corrected across the three within-language tests)

Language	n (academic/control)	Median (academic)	Median (control)	Cliff's δ	p (Holm)
Java	102 / 96	1.67	2.36	−0.41	< 0.001
JavaScript	93 / 95	2.47	3.08	−0.33	< 0.001
Python	69 / 72	3.20	4.53	−0.49	< 0.001

We also tested a second alternative explanation directly: that shorter-lived, less-maintained repositories, plausible for course projects that end when a semester does, rather than any academic/non-academic distinction, explain the lower complexity. Using each repository's GitHub-reported creation and last-push dates (523 of 549 repositories resolved; the rest lacked a retrievable date), academic repositories are, if anything, longer-lived than control repositories (median 103.6 days versus 55.2 days, Mann-Whitney $p < 0.001$), the opposite of what the maturity hypothesis predicts. An OLS regression of mean cyclomatic complexity on

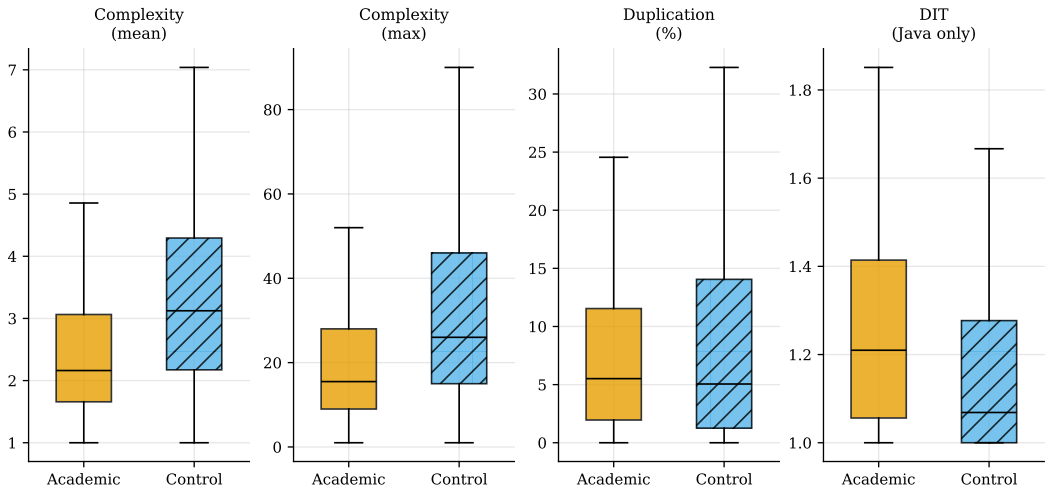


Figure 1: Distribution of key metrics, academic vs. control repositories

group membership and log-transformed repository lifespan leaves the group coefficient essentially unchanged from the unadjusted comparison (-0.997 unadjusted versus -0.975 adjusted, both $p < 0.001$) while lifespan itself is not a significant predictor ($p = 0.27$). Repository age does not explain the complexity difference.

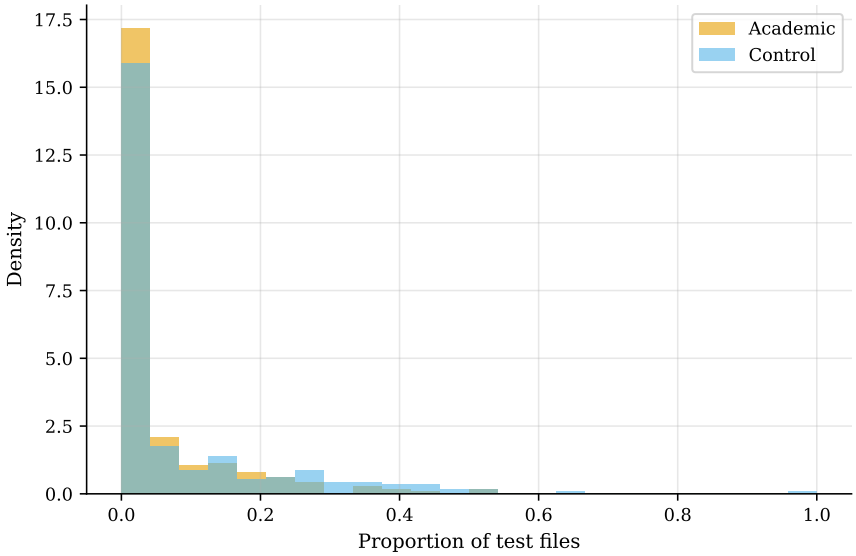


Figure 2: Distribution of the test-file-proportion proxy, academic vs. control

4.1 SonarQube corroboration

SonarQube's own complexity metrics corroborate the `lizard`-based finding across tool architectures: academic repositories have significantly lower complexity per function, both cyclomatic (median 2.07 vs. 2.98, $\delta = -0.458$, medium effect, $p < 10^{-14}$) and cognitive (median 1.31 vs. 3.00, $\delta = -0.426$, medium, $p < 10^{-12}$), the same direction and a comparable effect size to the `lizard`-based result in Table 3, now confirmed directly with SonarQube's reference analysis engine. `sqale_debt_ratio`, SonarQube's own technical-debt-ratio metric that our original substitution could not reproduce, is significantly *lower* in academic code ($\delta = -0.142$, $p = 0.025$), consistent with, not contradicting, the lower-complexity result. Duplication tells a more mixed story: SonarQube's duplicate-code detector finds significantly *higher* duplication in academic code (median 2.10% vs. 0.80%, $\delta = +0.181$, small effect, $p = 0.003$), the opposite direction from `jscpd`'s null result in Table 3 ($\delta \approx 0.03$); we treat this as a genuine tool-dependent discrepancy rather than resolve it in either direction (see Limitations). Code smells per 1,000 lines do not differ between groups ($p = 0.541$).

Figure 4 shows the ten most frequent SonarQube code-smell types, split by group. The single most common smell in both groups is a JavaScript/React-specific rule (*prop-types validation*), reflecting the language

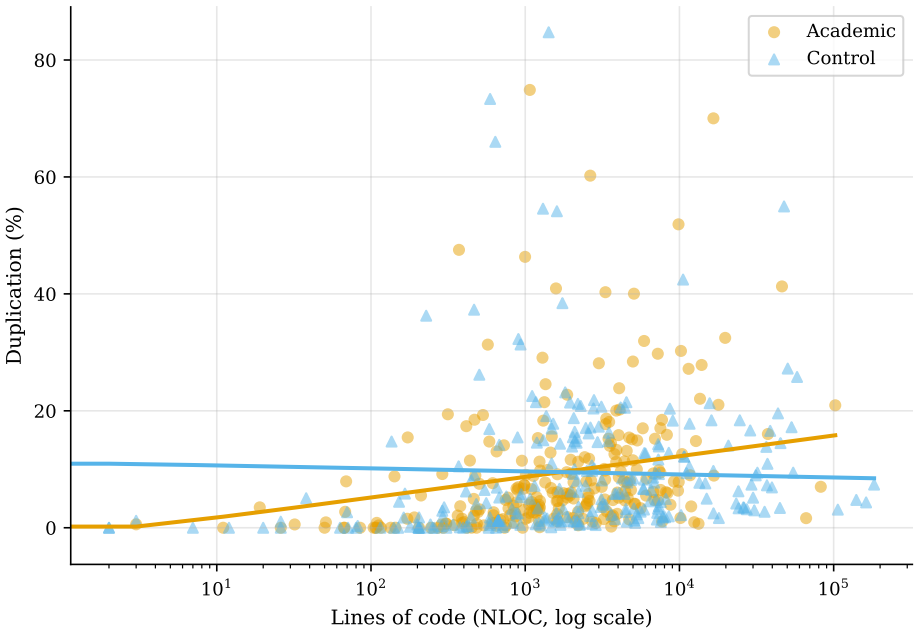


Figure 3: Repository size (lines of code) vs. code duplication, academic vs. control. Trend lines depict ordinary least squares (OLS) linear regressions for academic (solid) and control (dashed) repositories, with shaded 95% confidence bands

Table 5: Academic vs. control comparison on SonarQube-native metrics, 415 repositories (Holm-corrected)

Metric	Median (acad.)	Median (ctrl.)	Cliff's δ	Effect	p_{Holm}
Complexity / function (cyclomatic)	2.07	2.98	−0.458	medium	< 0.001
Complexity / function (cognitive)	1.31	3.00	−0.426	medium	< 0.001
Duplicated lines (%)	2.10	0.80	+0.181	small	0.003
`sqale_debt_ratio` (%)	0.50	0.70	−0.142	negligible	0.025
Code smells / 1,000 lines	27.2	23.3	+0.035	negligible	0.541

mix of the sample rather than an academic-specific pattern; academic code shows a markedly higher rate of two style-convention smells (`let`/`const` usage, package-naming convention), consistent with less mature project tooling and linting setup rather than with deeper structural problems. Not every smell in Figure 4 follows this direction, though: the default-unnamed-package smell (Java), also among the ten most frequent, is far more common in the *control* group (2,343 vs. 114 occurrences). This massive concentration in the control group likely originates from a few specific repositories with heavy code generation rather than a widespread practice, so we do not read the academic/control split as uniform across all ten smell types. Finally, we note that the SonarQube sub-sample suffers from differential attrition; see Limitations (§6) for a full accounting of its impact.

4.2 Self-admitted technical debt

As a complement to the structural metrics above, we scanned source comments in the same 415-repository SonarQube sub-sample for self-admitted technical debt (SATD) markers (`TODO`, `FIXME`, `HACK`, `XXX`, `BUG`, and literal mentions of "technical debt"), normalized per 1,000 lines of code. The median is zero in both groups (most repositories, academic and control alike, contain no SATD marker at all), but the distributions differ: a significantly larger share of academic repositories contain at least one SATD comment (42.2% vs. 23.0%), and academic repositories rank higher overall on this measure (Mann-Whitney $U = 25{,}349.5$, $p < 0.001$, Cliff's $\delta = +0.190$, small effect; IQR 0.00–0.52 academic vs. 0.00–0.00 control). This is driven by typicality rather than extremes: mean SATD density is actually higher in the control group (1.20 vs. 0.65 per 1,000 lines), because a handful of control repositories carry extreme outlier values (up to 120.9 per 1,000 lines) that Cliff's delta, an ordinal measure, is not sensitive to. Unlike the complexity result, this SATD finding runs *with* the original hypothesis: academic developers more often leave explicit, self-admitted markers of incomplete or provisional work in their code than the control group does, even though the code that remains

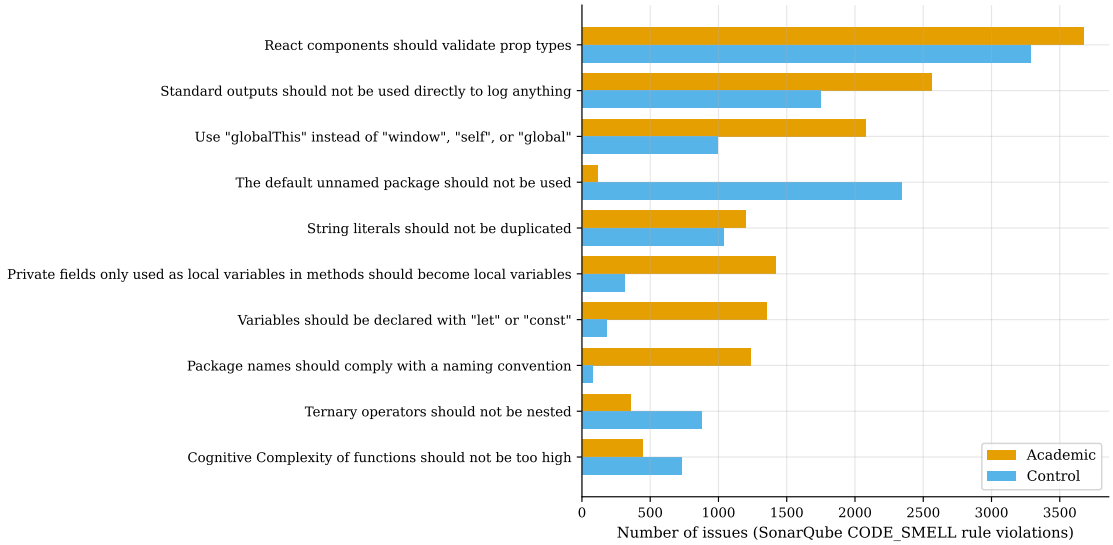


Figure 4: The ten most frequent SonarQube code-smell types, academic vs. control (415 repositories analyzed)

is structurally simpler by the metrics above.

5 Discussion

We report the complexity finding as it stands rather than reframing the paper around a different, more expected result: on this sample, academic Latin American repositories are structurally *simpler*, not more complex, than the size-matched control group. We consider the most plausible explanation to be a property of the control group rather than of the academic group: because repositories were matched to the academic sample's typically modest size, the control sample necessarily excludes large, mature, well-known open-source projects and instead draws from the same broad population of small, low-visibility GitHub repositories that academic work is drawn from: personal projects, bootcamp exercises, and other non-professional code that is not actually representative of production software engineering practice. If the "non-academic" baseline is itself closer to informal, learning-stage code than to professionally maintained software, the comparison this paper runs is better described as *labeled academic vs. unlabeled small repositories of unknown provenance* than as *academic vs. professional*, which would materially change what a lower complexity score in the academic group means: it may reflect course-scoped assignments that are deliberately narrow in requirements, rather than academic authors writing objectively better-structured code than typical small non-academic projects.

The DIT result, that academic code has measurably deeper inheritance hierarchies, is consistent with a specific, plausible mechanism: object-oriented design courses often assign or scaffold class-hierarchy exercises (e.g., requiring the use of inheritance to satisfy a rubric) in a way that unscaffolded personal projects do not, which would produce deeper hierarchies without those hierarchies reflecting either more sophisticated or more debt-laden design.

We take the near-zero test-file proportion in both groups, rather than only the academic group, at face value: it suggests that the near-absence of testing culture in this size class of repository is a broader phenomenon than an academic-specific one, at least by this proxy metric, and that interventions aimed at improving testing practice in student software may be addressing a gap that is not actually specific to the classroom.

For software engineering teaching in the region, the results point to specific course-level habits rather than to a general deficit in student code: inheritance used because an assignment requires it, almost no automated tests, and debt markers left in comments. Active-learning formats used in software engineering courses, among which a recent review in this journal finds games and project-based learning the most frequent, with team-based learning adopted in the authors' own department [19], are natural places to make design review, a minimum of automated tests, and the clean-up of `TODO`/`FIXME` markers explicit deliverables. Because many of these habits carry over into industry, work on training software development teams in the region, which adapts teaching strategies to the learning style most common among practising developers [20], is a complementary route for the same practices after graduation.

That the SonarQube sub-sample reproduces the complexity result across distinct parser implementations matters for how much weight the finding should carry: it is not an artifact of `lizard`'s particular parsing or counting conventions, since a structurally different tool, one that also computes a composite, remediation-cost-based debt ratio rather than a single structural proxy, converges on the same direction and a similar magnitude. The lower `sqale_debt_ratio` in academic code extends this rather than complicating it: SonarQube's debt ratio aggregates many rule violations, beyond just complexity, so a lower ratio despite a higher duplication rate suggests that whatever is driving academic code's higher SonarQube-detected duplication is not enough, on its own, to outweigh the complexity advantage in SonarQube's own composite accounting.

The duplication discrepancy between `jscpd` (null, default settings) and SonarQube's detector (small, significant, and in the direction the original hypothesis expected) is at least partly explained by the two tools' different default block-matching thresholds. We isolated this factor directly: re-running `jscpd` with `--min-tokens 100` (approximating SonarQube's typical Java threshold of roughly 100 tokens or 10 lines) on the identical 415-repository sub-sample used for the SonarQube comparison, holding the sample fixed so only the threshold changes, reverses the null result. At the higher threshold, `jscpd` finds significantly *higher* duplication in academic code (median 1.77% vs. 1.10%, Cliff's $\delta = +0.122$, small effect, $p = 0.031$), the same direction SonarQube reports, rather than the null result `jscpd` gives at its own default (roughly 50 tokens) on the larger 549-repository sample. This does not fully resolve the discrepancy, since the effect size at the matched threshold ($\delta = 0.122$) is still smaller than SonarQube's ($\delta = 0.181$) and the sample-size factor remains conflated with the original default-threshold null result, but it shows the threshold is a real, non-trivial contributor: a more permissive duplicate-detection threshold on the same tool, same sample, changes both the significance and the direction of the finding.

6 Limitations

The keyword-plus-contributor-location heuristic used to identify academic repositories is imperfect and can admit false positives (a repository containing the word "tesis" in an unrelated context) or miss true academic repositories that do not use any of the searched terms; we did not attempt manual verification at scale. The count of 5,418 keyword-matched candidates is the figure reported by our search script at collection time; the unfiltered candidate list was not archived, so that number cannot be re-derived from the released data, whereas the 300 accepted academic and 300 control repositories, and every metric computed on them, can. More generally, GitHub-mining studies are known to over-represent personal, inactive, and non-engineered projects unless explicitly filtered [8, 9]; our size threshold, minimum-commit filter, and (for the academic sample) contributor-location verification each narrow the sample somewhat in this direction, but we did not apply a dedicated engineered-project classifier, so some informal or abandoned repositories likely remain in both groups. The heuristic also identifies only GitHub repositories, excluding GitLab, institutional repositories, and any academic work never pushed to a public host at all, a substantial and likely non-random exclusion, since the most informal, least-polished student work is probably the least likely to be published publicly in the first place. As discussed above, the control group's size-matching means it should not be read as representative of professional software in general. Test coverage is a naming-convention proxy, not measured execution coverage. `ck`'s object-oriented metrics apply only to the Java subset of each sample (roughly one third), reducing statistical power for CBO, LCOM, DIT, and WMC relative to the language-agnostic metrics. The SonarQube robustness check (Table 5) covers a smaller, non-identical sub-sample (415 of the original 546 re-cloned repositories) because source code had to be re-cloned from GitHub after the original clones were deleted to save disk space, and because SonarQube's Java analyzer, run without compiled bytecode, fails outright on some repositories rather than degrading gracefully. This attrition is not evenly distributed across groups: 223 of 274 academic repositories (81.4%) completed analysis versus 192 of 272 control repositories (70.6%), an 11-percentage-point gap, and failures skewed toward genuinely large repositories, which are more common in the control group. We checked this directly rather than leaving it as an unknown: joining the primary sample's `lizard` complexity measurements to SonarQube-attrition status, attrited repositories are far larger than retained ones in both groups (median NLOC 5,497 vs. 1,229 academic, 7,709 vs. 1,369 control, $p < 0.001$ both), directly confirming size-driven analyzer failure as the mechanism. On complexity itself, attrited repositories trend higher than retained ones in the academic group (median `cc_mean` 2.35 vs. 2.15) and lower in the control group (2.66 vs. 3.23), a pattern that, if it held up statistically, would mean the SonarQube sub-sample loses relatively complex academic repositories and relatively simple control repositories, which would widen the apparent academic-versus-control complexity gap in Table 5 relative to the true populations, in the same direction as the paper's central finding. However, neither within-group difference reaches significance at these sample sizes (48 attrited academic repositories, $p = 0.34$; 78 attrited control repositories, $p = 0.12$), so this remains a suggestive but inconclusive pattern rather than a confirmed bias; we report it rather than the original blanket "we do not know" because a direction, even a

statistically uncertain one, is more informative than none, but we do not treat the SonarQube corroboration as unbiased in its own right. `sqale_debt_ratio` and the code-smell-type facets are SonarQube's default rule profile ("Sonar way"), not a profile tuned for either language mix or academic code specifically, and a different quality profile could shift the exact smell-type ranking in Figure 4 without necessarily changing the aggregate complexity and debt-ratio results. Repositories were matched between groups at the group level, by language and a size band, not paired one-to-one; Mann-Whitney U (an independent-samples test) is therefore the correct test for the design we ran, and we refer to the design as matched-sample rather than paired throughout. The self-admitted technical debt (SATD) measure is a simple keyword count over source comments, not a validated SATD classifier; it covers only the 415-repository SonarQube sub-sample (not the full 549-repository primary sample, for the same disk-space reason discussed above), and it does not distinguish a genuine debt marker from an unrelated use of a common word (e.g. `BUG` inside an unrelated identifier or string), so the reported counts should be read as a coarse proxy rather than a precise measurement. Lines of code is used both as a matching criterion (via the size band) and as one of the nine compared metrics (Table 3); its own comparison ($p = 0.322$, not significant after correction) should be read as a check that the size-band matching worked as intended, not as an independent finding about repository size.

7 Conclusion

In a matched-sample comparison of 274 Latin American academic repositories against 275 size- and language-matched non-academic repositories, academic code shows significantly *lower* cyclomatic complexity than the control group (the opposite of our initial hypothesis) while showing significantly deeper inheritance hierarchies, as hypothesized. Coupling, cohesion, duplication, and a test-coverage proxy show no significant difference after correction, and testing is near-absent in both groups rather than distinctively academic. We attribute the inverted complexity result chiefly to the composition of the size-matched control group, which is itself drawn from small, often informal repositories rather than mature open-source software, and recommend that future comparative studies of academic code quality either use a professionally curated control set or explicitly frame their comparison as academic-versus-small-repository rather than academic-versus-professional. A follow-up check with SonarQube Community itself, run without Docker on a 415-repository sub-sample after our original tool substitution, reproduces the direction of the lower-complexity finding using SonarQube's own complexity metric and shows academic code has a lower SonarQube technical-debt ratio; because SonarQube and our primary tool compute the same underlying cyclomatic-complexity construct, we treat this as a cross-tool consistency check rather than an independent replication, and the one clear point of disagreement across tools, duplication rate, is partly explained by the tools' different duplicate-detection thresholds: re-running `jscpd` with a SonarQube-matched threshold on the same 415-repository sub-sample reverses its null result to agree with SonarQube's direction (academic higher), though a smaller effect size than SonarQube's own, so the discrepancy is narrowed rather than fully resolved. On the same sub-sample, a self-admitted technical debt (SATD) comment count runs in the direction the original hypothesis expected, unlike the structural metrics: academic repositories more often contain explicit `TODO`/`FIXME`-style markers of unfinished work (Cliff's $\delta = +0.190$, $p < 0.001$), even though the structural complexity of that code is lower.

Data Availability

The repository lists, extracted metrics, and analysis code are permanently archived and openly accessible in Zenodo at `https://doi.org/10.5281/zenodo.22907773` and GitHub at `https://github.com/JuanAvilesP3/cleiej-deuda-tecnica`. Cloned source code itself is not redistributed; only derived metrics and repository identifiers are shared, consistent with each source repository's own license.

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Conflict of Interest

The authors declare that they have no conflict of interest.

Use of Generative AI

The authors used generative AI tools (large language models) as an assistant for language editing, support in writing and debugging analysis code, and LaTeX formatting. The research questions, study design, data collection, execution of the analyses, results, and their interpretation are the authors' own work; all AI-assisted output was reviewed and verified by the authors, who take full responsibility for the content of this article. No figure was produced with image-generation tools.

Author Contributions

Isaac David Torres-Paredes: Conceptualization, Methodology, Project Administration, Supervision, Writing - Original Draft. **Juan Pablo Aviles-Esparza:** Software, Formal Analysis, Data Curation, Visualization, Writing - Review & Editing. **Italo Javier Tenempaguay-Granizo:** Validation, Investigation, Writing - Review & Editing.

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